# DengueAI Pro: AI-Augmented Dengue Risk Intelligence System

## Submitted to

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## Team

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## Track

HealthTech (Track 3)

## Sustainable Development Goal

SDG 03 – Good Health and Well-Being

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# Executive Summary

Dengue remains one of the most serious public health challenges in Bangladesh. Despite decades of control efforts, the country continues to experience severe outbreaks, resulting in thousands of hospitalizations and deaths every year. The primary weakness of the current system is that it is reactive rather than preventive.

DengueAI Pro is an AI-powered public health intelligence platform designed to transform dengue management from reaction to prevention. The platform combines machine learning, satellite imagery analysis, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Graph Databases to predict dengue risks, detect mosquito breeding sites, support authorities with intelligent recommendations, and engage citizens in prevention activities.

The system aims to reduce outbreak severity, improve resource allocation, and create a scalable model that can be deployed across Bangladesh and other tropical countries.

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# 1. Problem Statement

Bangladesh has experienced increasingly severe dengue outbreaks over the last decade. In 2023 alone:

- 321,179 people were hospitalized.
- 1,705 people died.
- Approximately 40–50 million urban residents remained at risk.

The current dengue response system suffers from five major weaknesses:

1. Manual mosquito breeding site detection.
2. Lack of weather-based outbreak forecasting.
3. No data-driven resource prioritization.
4. Limited citizen participation.
5. Absence of a centralized outbreak intelligence system.

As a result, authorities often respond only after outbreaks have already begun.

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# 2. Proposed Solution: DengueAI Pro

DengueAI Pro is a five-layer AI-native platform that provides predictive, data-driven dengue prevention.

## Layer 1: Predictive Risk Engine

The prediction module analyzes weather and environmental data using:

- LSTM Neural Networks
- Random Forest Models

The system generates a SWAPI (Smart Ward Alert Prediction Index) score for each ward, indicating the likelihood of dengue outbreaks in the coming days.

## Layer 2: AI Water Detection

Potential mosquito breeding sites are detected through:

- Sentinel-2 Satellite Imagery
- U-Net Segmentation Models
- YOLOv11 Image Detection

Citizens can also submit photos through the mobile application, allowing the system to verify stagnant water locations automatically.

## Layer 3: LLM Dispatch Advisor

Large Language Models such as Claude and Gemini generate actionable recommendations for field officers.

The system provides:

- Priority locations
- Recommended interventions
- Resource allocation guidance
- Operational instructions

All recommendations are grounded using RAG to ensure reliability and reduce hallucinations.

## Layer 4: Graph Intelligence

A Neo4j Graph Database stores relationships between:

- Weather patterns
- Historical outbreaks
- Locations
- Mosquito breeding sites
- Intervention outcomes

This allows advanced outbreak reasoning and pattern discovery.

## Layer 5: Citizen and Authority Platform

A Bangla-first web and mobile platform enables:

- Citizen reporting
- SMS alerts
- Risk notifications
- Community engagement
- Authority dashboards

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# 3. System Workflow

The DengueAI Pro workflow consists of seven major stages:

### Step 1: Weather Collection

Weather data is collected every six hours from meteorological sources.

### Step 2: Risk Prediction

LSTM and Random Forest models calculate ward-level SWAPI scores.

### Step 3: Knowledge Retrieval

Historical outbreak information is retrieved from the RAG database and GraphDB.

### Step 4: Satellite Activation

High-risk areas trigger satellite image analysis.

### Step 5: Water Detection

AI models identify potential mosquito breeding sites.

### Step 6: Intelligent Dispatch

LLMs generate operational recommendations for field teams.

### Step 7: Continuous Learning

Feedback from field operations is used to retrain models and improve future predictions.

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# 4. Technology Stack

## Prediction Layer

- Python
- TensorFlow
- Scikit-Learn
- LSTM
- Random Forest

## Detection Layer

- U-Net
- YOLOv11
- OpenCV

- Sentinel-2
- Google Earth Engine

## AI Layer

- Claude
- Gemini
- LangChain
- RAG Pipeline
- PGVector

## Knowledge Layer

- Neo4j Graph Database
- LangGraph

## Platform Layer

- FastAPI
- PostgreSQL
- PostGIS
- React.js
- Redis

## DevOps

- Docker
- GitHub Actions
- AWS/GCP
- MLflow

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# 5. Business Model

## Revenue Streams

### City Corporation SaaS License

Annual subscription for city corporations.

### National Government Contracts

Surveillance dashboards and API access for public health agencies.

### SWAPI Open API

Usage-based API access for NGOs and research organizations.

### International Grants

Funding partnerships with organizations such as WHO and UNICEF.

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# 6. Market Opportunity

## Total Addressable Market (TAM)

$2.4 Billion

Vector-borne disease management market across South and Southeast Asia.

## Serviceable Available Market (SAM)

$180 Million

Bangladesh public health technology sector.

## Serviceable Obtainable Market (SOM)

$4.2 Million

Initial focus on Dhaka city corporations.

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# 7. Scalability Roadmap

## Phase 1 (0–6 Months)

- Pilot deployment in 10 wards of Dhaka.
- Citizen onboarding.
- Live SWAPI dashboard.

## Phase 2 (6–18 Months)

- Expansion to all 54 Dhaka wards.
- NGO API access.
- IoT sensor integration.

## Phase 3 (18–36 Months)

- Expansion to major Bangladeshi cities.
- Support for Chikungunya and Zika monitoring.

## Phase 4 (36+ Months)

- Regional deployment across South and Southeast Asia.
- Government licensing model.
- International partnerships.

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# 8. Expected Impact

DengueAI Pro aims to:

- Predict outbreaks before they occur.
- Reduce dengue-related hospitalizations and deaths.
- Improve efficiency of public health interventions.
- Increase citizen participation.
- Enable data-driven decision making.
- Strengthen Bangladesh's public health resilience.

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# Conclusion

DengueAI Pro introduces a proactive and intelligent approach to dengue prevention by integrating artificial intelligence, satellite monitoring, graph intelligence, and large language models into a single platform. Rather than reacting after outbreaks begin, the system empowers authorities and communities to take preventive action before mosquito populations and infections spread.

With its scalable architecture, strong business model, and alignment with SDG 03, DengueAI Pro has the potential to become a national public health infrastructure and a globally exportable HealthTech solution.